

28-2 ACL Troubleshooting - Answer Key

In this lab you will troubleshoot Access Control Lists.

ACL Troubleshooting

The previous network administrator was tasked with securing the network as follows:

1. PC1 should be the only PC which can Telnet to any router. (The network managers are not concerned about securing Telnet from one router to another).
2. Ping should not work to R2.

The security configuration has been tested and management found the following problems:

1. PC1 cannot Telnet to any router. PC3 can Telnet to any router.
2. R1 can ping R2.

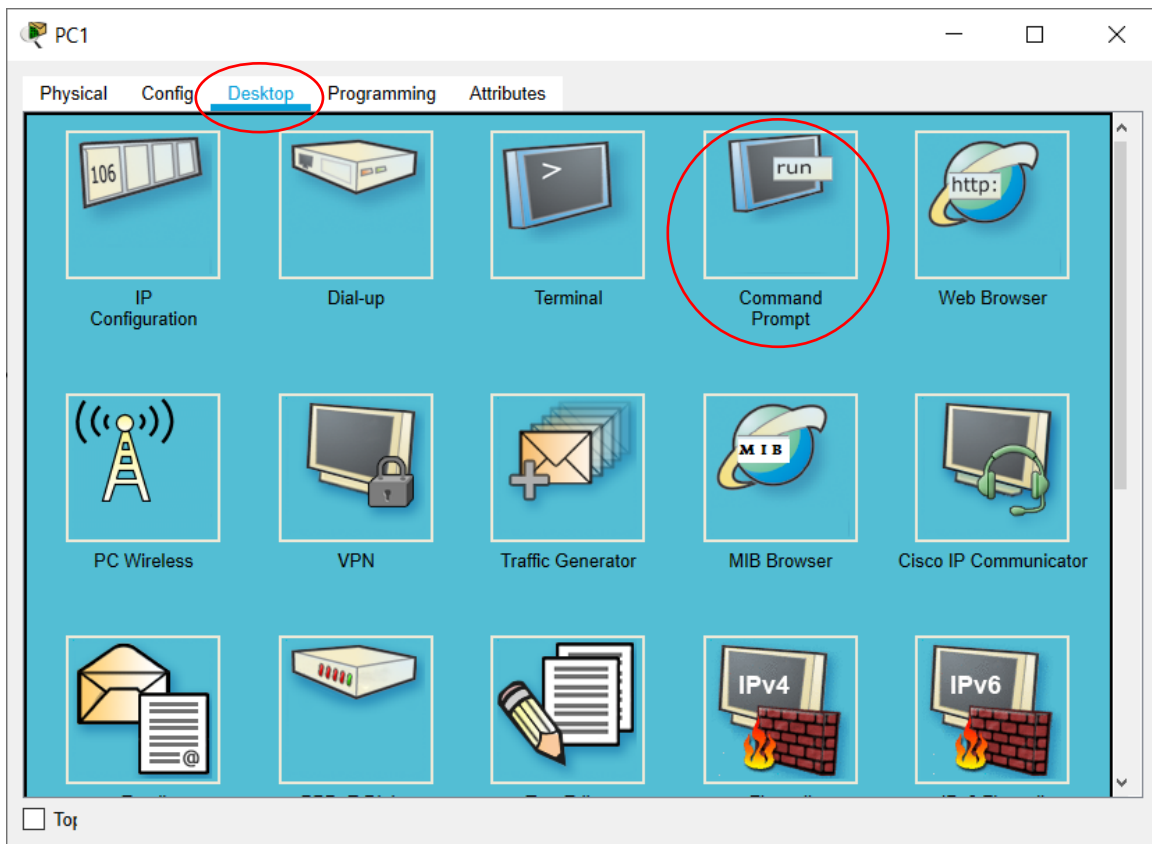
You have been brought in to help. View the configuration on R1 and R2 and then fix both problems.

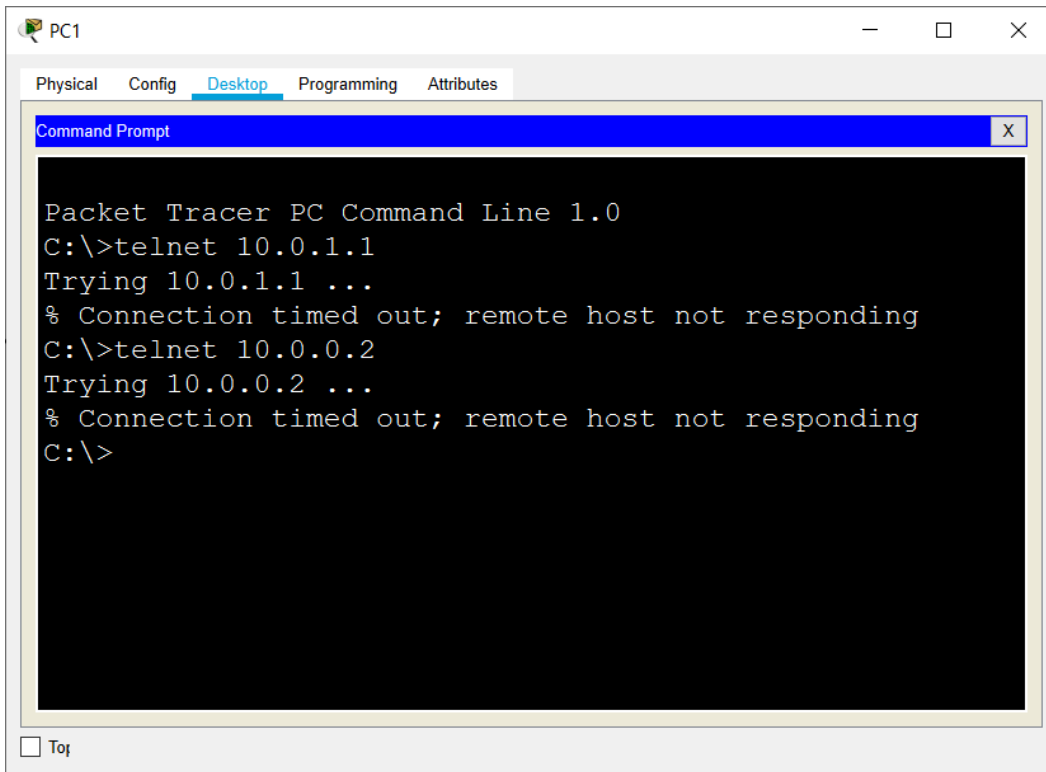
The Telnet password is **Flackbox**

1. PC1 cannot Telnet to any router. PC3 can Telnet to any router.

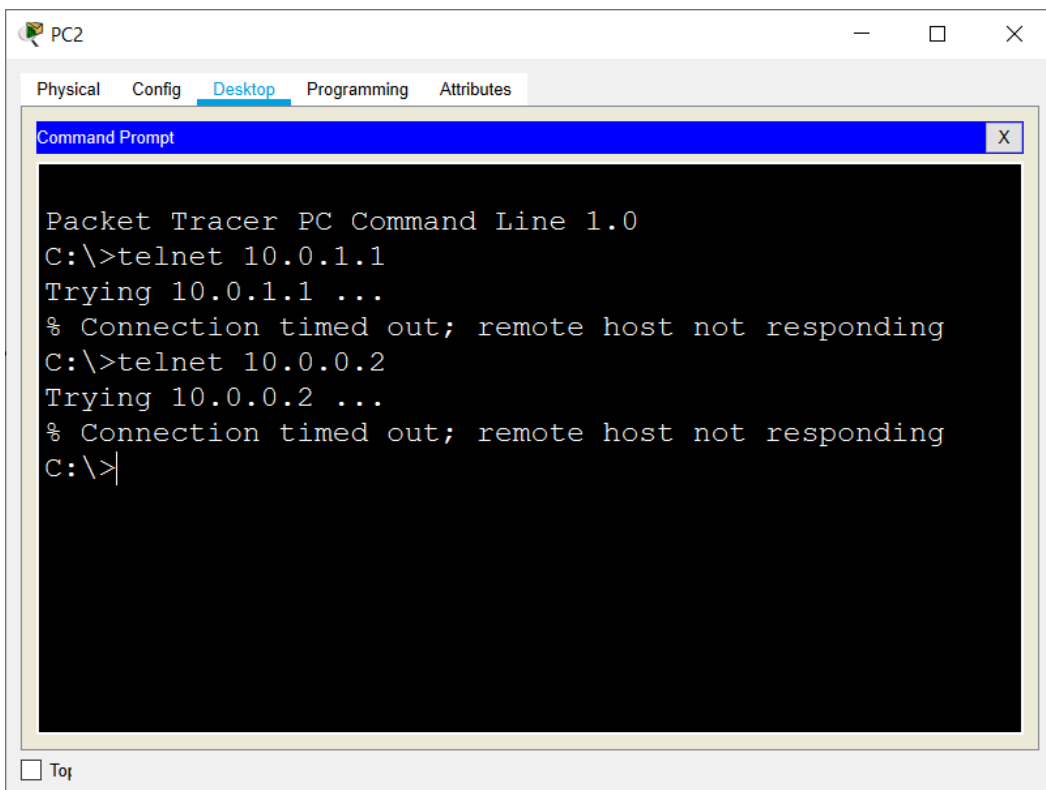
The first thing to do is verify the problem.

On PC1, open a Command Prompt and attempt to Telnet to 10.0.1.1 and 10.0.0.2

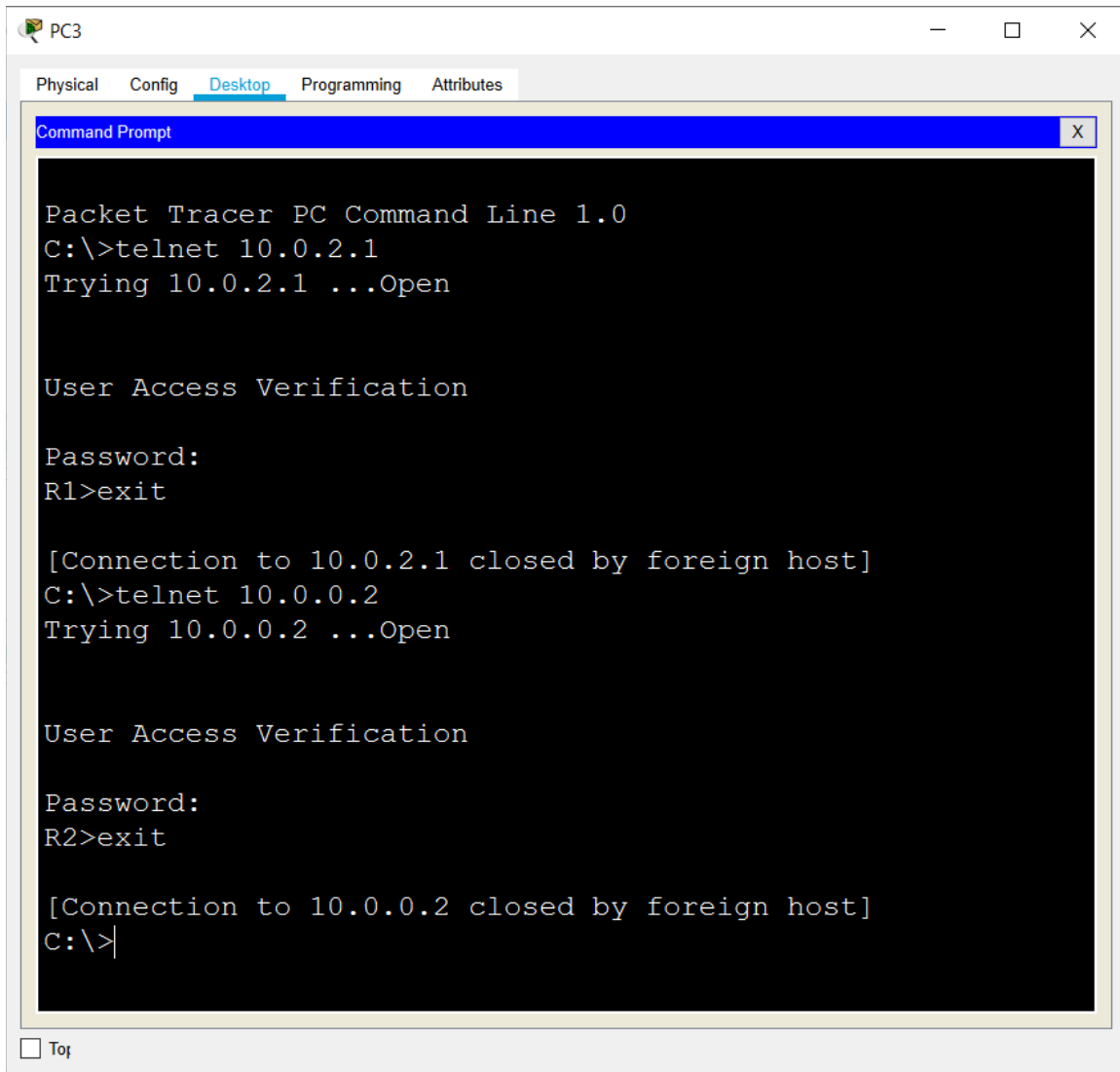




Also attempt to Telnet to 10.0.1.1 and 10.0.0.2 from PC2



On PC3, attempt to Telnet to 10.0.2.1 and 10.0.0.2



The screenshot shows a window titled 'PC3' with tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The Command Prompt shows the following text:

```
Packet Tracer PC Command Line 1.0
C:\>telnet 10.0.2.1
Trying 10.0.2.1 ...Open

User Access Verification

Password:
R1>exit

[Connection to 10.0.2.1 closed by foreign host]
C:\>telnet 10.0.0.2
Trying 10.0.0.2 ...Open

User Access Verification

Password:
R2>exit

[Connection to 10.0.0.2 closed by foreign host]
C:\>|
```

At the bottom of the Command Prompt window, there is a checkbox labeled 'Toq' which is currently unchecked.

This is the ACL configuration on R1 which is controlling the Telnet traffic:

```
R1#sh access-list
Extended IP access list Control_Telnet
 10 deny tcp 10.0.1.0 0.0.0.255 gt 1023 any eq telnet
 20 permit tcp host 10.0.1.10 gt 1023 any eq telnet
 30 permit ip any any
```

```
R1#show run
```

```
interface FastEthernet1/0
 ip address 10.0.1.1 255.255.255.0
 ip access-group Control_Telnet in
 duplex full
```

ACL Control_Telnet has been applied inbound on F1/0.

There are actually two problems here. The first is that neither PC1 nor PC2 can Telnet to R1 or R2, but PC1 should be able to.

The problem is that the ACE's have been entered in the wrong order:

```
10 deny tcp 10.0.1.0 0.0.0.255 gt 1023 any eq telnet
20 permit tcp host 10.0.1.10 gt 1023 any eq telnet
```

Entry 10 is denying Telnet access from all hosts in the 10.0.1.0/24 subnet, including PC1. The router stops processing the ACL as soon as there is a match so the Telnet traffic from PC1 is denied, the ACL never gets as far as entry 20.

The source port in the ACEs isn't doing any harm but it's not needed so we can remove that.

Apply a corrected ACL to interface F1/0. It's easiest to create a new ACL to replace the old one.

```
R1(config)#ip access-list extended Telnet_Control_F1/0
R1(config-ext-nacl)#permit tcp host 10.0.1.10 any eq telnet
R1(config-ext-nacl)#deny tcp 10.0.1.0 0.0.0.255 any eq
telnet
R1(config-ext-nacl)#permit ip any any
```

Don't forget to apply it to the interface.

```
R1(config)#int f1/0
R1(config-if)#ip access-group Telnet_Control_F1/0 in
```

Delete the old ACL to prevent confusion.

```
R1(config)#no ip access-list extended Control_Telnet
```

The second problem is that no ACL is applied inbound on interface F0/1 to deny Telnet access from hosts in the 10.0.2.0/24 subnet.

Configure the ACL.

```
R1(config)#ip access-list extended Telnet_Control_F0/1
R1(config-ext-nacl)#deny tcp any any eq telnet
R1(config-ext-nacl)#permit ip any any
```

Apply it to the interface.

```
R1(config)#int f0/1
R1(config-if)#ip access-group Telnet_Control_F0/1 in
```

Verify the problem is fixed. PC1 should be able to Telnet to both R1 and R2. The other PCs should not be able to Telnet to either router.

```
PC1#telnet 10.0.1.1
Trying 10.0.1.1 ... Open
User Access Verification
Password:
R1>exit
[Connection to 10.0.1.1 closed by foreign host]
```

```
PC1#telnet 10.0.0.2
Trying 10.0.0.2 ... Open
User Access Verification
Password:
R2>exit
[Connection to 10.0.0.2 closed by foreign host]
```

```
PC2#telnet 10.0.1.1
Trying 10.0.1.1 ...
% Connection timed out; remote host not responding
```

```
PC2#telnet 10.0.0.2
Trying 10.0.0.2 ...
% Connection timed out; remote host not responding
```

```
PC3#telnet 10.0.1.1
Trying 10.0.1.1 ...
% Connection timed out; remote host not responding
```

```
PC3#telnet 10.0.0.2
Trying 10.0.0.2 ...
% Connection timed out; remote host not responding
```

2. R1 can ping R2. Verify the problem.

As desired, PC1 cannot ping R2:

```
C:\>ping 10.0.0.2
```

Pinging 10.0.0.2 with 32 bytes of data:

```
Reply from 10.0.1.1: Destination host unreachable.  
Reply from 10.0.1.1: Destination host unreachable.  
Reply from 10.0.1.1: Destination host unreachable.  
Reply from 10.0.1.1: Destination host unreachable.
```

Ping statistics for 10.0.0.2:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

As desired, PC3 also cannot ping R2:

```
C:\>ping 10.0.0.2
```

Pinging 10.0.0.2 with 32 bytes of data:

```
Reply from 10.0.2.1: Destination host unreachable.  
Reply from 10.0.2.1: Destination host unreachable.  
Reply from 10.0.2.1: Destination host unreachable.  
Reply from 10.0.2.1: Destination host unreachable.
```

Ping statistics for 10.0.0.2:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

R1 can ping R2, this is not the desired behaviour:

```
R1#ping 10.0.0.2
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/20/20 ms

This is the ACL configuration which is controlling ping on R1:

```
R1#sh access-list
Extended IP access list 100
    10 deny icmp any host 10.0.0.2 (10 matches)
    20 permit ip any any (20 matches)
```

```
R1#sh run
```

```
interface FastEthernet0/0
 ip address 10.0.0.1 255.255.255.0
 ip access-group 100 out
 duplex auto
```

The problem is that ACLs do not apply to traffic which is originated on the router itself. To deny Ping traffic into R2, the ACL must be configured and applied on R2. I'll use a named access list but numbered is also fine.

```
R2(config)#ip access-list extended Control_ICMP
R2(config-ext-nacl)#deny icmp any host 10.0.0.2
R2(config-ext-nacl)#permit ip any any
```

```
R2(config)#int f0/0
R2(config-if)#ip access-group Control_ICMP in
```

Remove the old ACL on R1 to avoid confusion.

```
R1(config)#no access-list 100
```

```
R1(config)#int f0/0
R1(config-if)#no ip access-group 100 out
```

Verify none of the PCs or R1 can ping R2.

PC1 cannot ping R2:

```
C:\>ping 10.0.0.2
```

Pinging 10.0.0.2 with 32 bytes of data:

```
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
```

Ping statistics for 10.0.0.2:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

PC3 also cannot ping R2:

```
C:\>ping 10.0.0.2
```

Pinging 10.0.0.2 with 32 bytes of data:

```
Reply from 10.0.2.1: Destination host unreachable.  
Reply from 10.0.2.1: Destination host unreachable.  
Reply from 10.0.2.1: Destination host unreachable.  
Reply from 10.0.2.1: Destination host unreachable.
```

Ping statistics for 10.0.0.2:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

R1 now also cannot ping R2:

```
R1#ping 10.0.0.2
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:

UUUUU

Success rate is 0 percent (0/5)